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P8 · Chatter: Detecting social interaction in a room with non-visual modalities

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Question: **can three phone microphones locate a speaker in a room?**

Why it matters

Our goal is detecting conversation groups from audio only, on-device, with no camera. Knowing where each speaker stands is the first step.

Setting and constraint

One indoor room. One person walking and reading aloud. Three separate single-mic phones on the floor. Constraint: localization accuracy.

Success criterion

Median error ≤ 50 cm on a held-out recording, clearly below the no-audio baseline (98 cm).

Status: everything shown is offline replay of three recorded sessions. No live runs or simulation yet.

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Approach: phone spectra → small model → position

AUDIO PATH: WHAT THE SYSTEM USES

3 phones
2 iPhones, 1 iPad; synced by claps

→

1 s windows
16 kHz, 300–4000 Hz

→

Spectral features
8 bands or log-mel, relative to phone mean

→

Small MLP / CNN
6–12k parameters

→

Position
(x, y), or one 50 cm zone

GROUND-TRUTH LABELS ONLY: CAMERA NEVER USED AT INFERENCE

Wall camera
1080p, 20 fps

→

YOLO11m-pose
Reused: ankle pixels

→

Our calibration
1.6 cm error on measured distances

→

Floor position
Label for every window

Works and tested
Tracker, sync, dataset (3 sessions, 2,963 windows), GCC-PHAT baseline, learned models

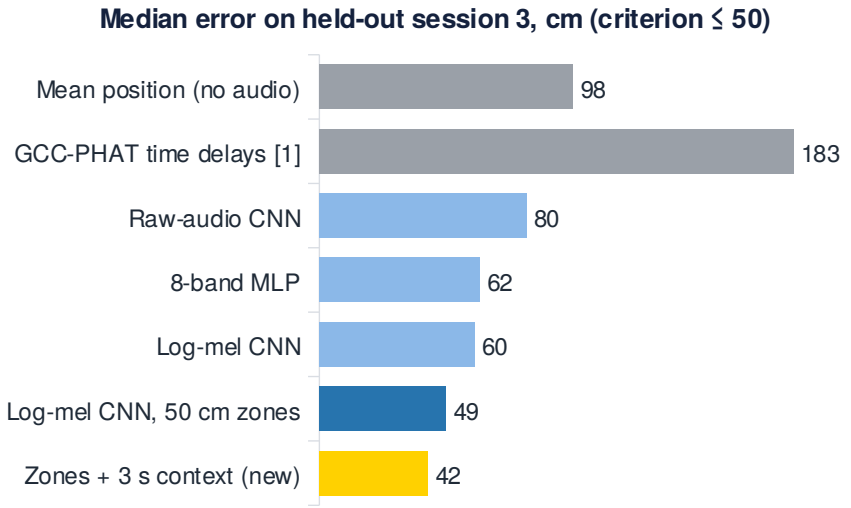
Mode: recorded-data replay
All results offline on recordings; no live inference or simulation yet

Planned
Several speakers, speaker counting, on-device deployment, live runs

Built by us: calibration, tracker pipeline, sync, dataset, all models and baselines. Reused: YOLO11m-pose (Ultralytics), PyTorch, SciPy.

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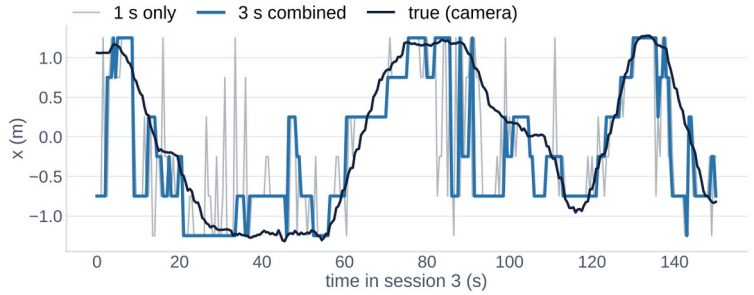
Evidence: 49 cm, and 42 cm with 3 s of context



NEW SINCE NOV 2: COMBINE 3 s OF ANSWERS

49 → 42 cm gain 95% CI 4–11 cm

Past-only (live-usable): 46 cm, smaller gain



Log-mel CNN, first 150 s of session 3: jumps removed

Supports: position cues live in the frequency balance between phones; timing fails.

Not shown: other rooms; unwalked floor (82–90 cm); several speakers.

Main uncertainty: combining assumes one continuous speaker, as recorded; turn-taking and overlap are untested.

Preliminary, measured on recorded-data replay: train sessions 1–2, test once on session 3 (826 windows). CI: block bootstrap, 30 s blocks. Code: github.com/ZiangXue/chatter/tree/c2600e6/feasibility/experiment

[1] Knapp & Carter, "The generalized correlation method for estimation of time delay," IEEE Trans. ASSP, 1976.

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Plan: localize and count speakers, then groups

Evaluation	Plan
Baselines	Mean position / majority zone; GCC-PHAT
Primary metric	Median localization error (cm), held-out session
Test data	New recordings covering unwalked floor area
Stress / ablation	Two overlapping speakers; drop one phone
Missing	Labelled multi-speaker data; consent to record others

Minimum deliverable
Audio-only localization and counting of 1–2 speakers

Largest risk
Overlapping speech breaks the spectral cue

Fallback
Add UWB tags as a second modality

NEXT TASKS (OWNER, TARGET DATE)

Record unwalked floor area, claps at both ends; check tracker while walking
Joseph · by Nov 14

Ablations: temporal smoothing done (Nov 6); next, drop one phone
Ziang · by Nov 13

Two-speaker test: simulate, then record
Both · by Nov 19

Our question for you: Is ~50 cm localization enough to pursue conversation-group detection, or should we add UWB now?